Collatz Dynamics as Arithmetic Topology: A Framework Connecting 2-Adic Transfer Operators to Kim's Arithmetic Chern–Simons Theory

JGPTech
github.com/JGPTech/Fun

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Abstract

We propose a structural framework connecting the 2-adic dynamics of the Collatz map to the arithmetic topology program and to Minhyong Kim's arithmetic Chern–Simons theory. Viewing the shortened Collatz map as a transfer operator on 2-adic probability measures, we show that its trajectories are naturally interpreted as paths in the 2-adic tree, and we introduce a candidate scalar shadow $\mathrm{hol}(\gamma_n) = \sum_k v_2(\gamma_n^{(k)})$ of an abelianized $\mathbb{Z}_p$-valued holonomy associated to each finite trajectory. Empirical computation for $n \leq 100$ shows that this quantity is approximately linear in trajectory length (Pearson correlation $r = 0.96$), with normalized mean ≈ 1.22, broadly consistent with the uniform-bit prediction $\mathbb{E}[v_2] = 1$. The Orzechowski threshold $M_2 = 4$ is located empirically at 96.2% of total collapse time and coincides with a sharp drop in the v_2-mean ($\Delta\bar{v}_2 = -0.48$), consistent with its proposed role as a ramification boundary. The supermartingale conjecture for the expectation $\mathbb{E}_{\mu_k}[|\cdot|_2]$ is supported across $N \in \{50, 100, 200, 500, 1000, 2000\}$ with $\sum_k \varepsilon_k$ uniformly bounded. We outline the open problems required to lift the scalar shadow to Kim's $\mathbb{Z}_p$-valued invariant and to formalize the proposed correspondence row by row. This positions Collatz dynamics as a testbed for arithmetic path invariants in a fully explicit discrete setting.

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1 Introduction

The framework developed here builds on several independent lines of work, each of which contributes a specific structural ingredient.

Mazur's 1964 observation [1] that prime ideals in Spec($\mathbb{Z}$) behave formally like knots in a 3-manifold initiated what is now called arithmetic topology. The dictionary was developed systematically by Morishita [5], Kapranov [2], and Reznikov [3], and we refer to it throughout as the M^2KR dictionary. Kim's arithmetic Chern–Simons theory [6] provides the cohomological machinery currently used to assign holonomy and linking invariants to paths in arithmetic fundamental groups, and is the natural target framework for any path invariant arising from a discrete dynamical system on $\mathbb{Z}_p$. Siegel's (p, q)-adic spectral framework [9] establishes that the shortened Collatz map is conjugate to the 2-adic shift on $\mathbb{Z}_2$, which is what makes Collatz trajectories paths in the appropriate sense. Orzechowski's recent theorem [10] on minimal paradoxical decompositions of non-Archimedean spaces provides the boundary condition we identify with the loss of structural richness in the Collatz ensemble. The empirical 2-adic transfer operator framework on which we build the holonomy was developed in [11].

The contribution of the present paper is to assemble these ingredients into a single working framework. Specifically, we propose that Collatz trajectories provide a concrete and computable family of paths on which Kim-type holonomy invariants may be defined and studied, we define a candidate scalar shadow of such an invariant and compute it empirically, and we identify the Orzechowski threshold in the Collatz system with the ramification boundary on the arithmetic side. The framework is laid out end to end; steps requiring original proof are collected in Section 7.

2 The Collatz Framework on $\mathbb{Z}_2$

2.1 The Shortened Map and Its Transfer Operator

Throughout we work with the shortened Collatz map

$$T_3(n) = \begin{cases} n/2 & \text{if } n \text{ even,} \\ (3n+1)/2 & \text{if } n \text{ odd.} \end{cases}$$

A finite trajectory starting at n is the sequence $\gamma_n = (n, T_3(n), T_3^2(n), \ldots, T_3^{|\gamma_n|}(n))$, terminating when the iterate enters the attractor cycle $\{1, 2\}$.

For $N \geq 1$, set $S_0 = \{1, \ldots, N\}$ and let μ_0 be the uniform measure on S_0. Define the measure evolution by pushforward

$$\mu_{k+1} = (T_3)_* \mu_k, \qquad S_{k+1} = T_3(S_k).$$

The observable of primary interest is the 2-adic absolute value $f(n) = |n|_2 = 2^{-v_2(n)}$, where $v_2(n) = \max\{j \geq 0 : 2^j \mid n\}$. Its expectation under μ_k is

$$\mathbb{E}_{\mu_k}[f] = \frac{1}{|S_k|} \sum_{n \in S_k} 2^{-v_2(n)}.$$

2.2 The Supermartingale Conjecture

Conjecture 2.1 (Supermartingale property, [11])**.** *$\mathbb{E}_{\mu_k}[f]$ is a supermartingale up to bounded fluctuations with respect to the natural filtration $\mathcal{F}_k = \sigma(S_0, \ldots, S_k)$:*

$$\mathbb{E}\big[\mathbb{E}_{\mu_{k+1}}[f] \,\big|\, \mathcal{F}_k\big] \leq \mathbb{E}_{\mu_k}[f] + \varepsilon_k,$$

with $\varepsilon_k \geq 0$, ε_k bounded, and $\sum_k \varepsilon_k < \infty$ (in expectation under the ensemble measure).

Empirical evidence across six scales of N is collected in Section 6 and supports the conjecture in its Cesàro form; step-by-step monotonicity is not asserted.

2.3 Vacuum and the Orzechowski Boundary

The *vacuum* is the asymptotic regime in which the parity branching variable $B(n) = \mathbf{1}_{n \text{ odd}}$ becomes independent of f under μ_k. The terminology is heuristic and reflects statistical independence rather than a physical vacuum. Following [11], vacuum onset is detected by five measures $M_1, \ldots, M_5$, the most structurally significant of which is the image-diversity measure $M_2 = |T_3(S_k)|$.

The threshold $M_2 \geq 4$ is motivated by Orzechowski's theorem [10]: any field with a nontrivial non-Archimedean valuation admits a paradoxical decomposition into exactly four pieces under its affine isometry group, and four is the minimum. The Collatz vacuum sits at the boundary at which the ensemble loses the structural richness required for such a decomposition.

2.4 The 2-Adic Conjugacy

Siegel [9] establishes that T_3 is topologically and measurably conjugate to the 2-adic shift σ on $\mathbb{Z}_2$. The conjugacy does not preserve finitely supported measures: in particular, it does not preserve collapse times or the distributional properties of finite ensembles, since collapse times for T_3 on $\{1, \ldots, N\}$ are observed empirically to scale roughly as $O(\log N)$ while orbits of σ on truncated 2-adic ensembles scale as $O(N^c)$. What the conjugacy does provide is the identification of Collatz trajectories with genuine paths in $\mathbb{Z}_2 = \varprojlim \mathbb{Z}/2^n\mathbb{Z}$, which is the setting in which Kim's theory is formulated.

3 The M^{2}KR Dictionary and Kim's Theory

3.1 The Core Analogy

The foundational correspondence of arithmetic topology asserts:

3-dimensional topology	**Algebraic number theory**
Knot $K \subset S^3$	Prime $\mathfrak{p} \subset \mathrm{Spec}(\mathbb{Z})$
3-manifold M	Ring of integers $\mathcal{O}_K$
Knot complement $S^3 \setminus K$	$\mathrm{Spec}(\mathbb{Z}) \setminus \{\mathfrak{p}\}$
Linking number	Legendre symbol
Alexander polynomial	Iwasawa polynomial
Milnor $\bar{\mu}$-invariants	Multiple residue symbols
Knot group $\pi_1(S^3 \setminus K)$	Galois group with restricted ramification

The dictionary is grounded in the étale-topological interpretation of $\mathrm{Spec}(\mathbb{F}_p) \hookrightarrow \mathrm{Spec}(\mathbb{Z})$ as a knot-like inclusion. Of the three layers of formalization, only the first — the Alexander/Iwasawa correspondence — is currently a theorem; Kim's program represents the active formalization of the second layer.

3.2 Kim's Arithmetic Chern–Simons Theory

Kim's framework [6] constructs a Chern–Simons-type functional in the arithmetic setting, with $\mathrm{Spec}(\mathcal{O}_K)$ playing the role of the 3-manifold and p-adic Galois representations playing the role of gauge connections. The central technical object is the unipotent fundamental group $\pi_1^{\mathrm{un}}(\mathrm{Spec}(\mathbb{Z}) \setminus S, b)$ for a basepoint b and a finite set of primes S. Paths in this space carry holonomy data valued in a p-adic Lie group. Recent work [8] connects this framework to p-adic L-functions via path integrals, and [7] treats entanglement entropies in the abelian case.

4 The Bridge: Collatz Trajectories as Arithmetic Paths

4.1 Structural Observation

A Collatz trajectory $(n, T_3(n), T_3^2(n), \ldots)$ is a sequence of positive integers and, via the embedding $\mathbb{N} \hookrightarrow \mathbb{Z}_2$, a path in the 2-adic tree. Since $\mathrm{Spec}(\mathbb{Z}_2)$ is a natural subspace of the arithmetic setting in which Kim's theory is formulated, Collatz trajectories provide a concrete family of paths to which that theory may be meaningfully applied. The 2-adic conjugacy of [9] ensures this identification respects the natural measurable structure.

4.2 A Three-Way Translation Table

The following table proposes the correspondence between the Collatz, $\mathrm{M}^2\mathrm{KR}$, and Kim layers. Verification status applies to the Collatz column specifically: **(V)** indicates the topology–arithmetic correspondence is directly verified against Morishita [4] Section 1 *and* the Collatz identification is essentially tautological; **(V/P)** indicates the topology–arithmetic correspondence is verified but the Collatz identification is proposed and interpretive; **(P)** indicates a row original to this work.

Collatz	**Topology**	**Arithmetic**	
Trajectory $\gamma_n = \{T_3^k(n)\}$ as path in $\mathbb{Z}_2$	Knot $K \hookrightarrow S^3$, $S^1 \hookrightarrow M$	Prime $\mathrm{Spec}(\mathbb{F}_\mathfrak{p}) \hookrightarrow \mathrm{Spec}(\mathcal{O}_k)$	**(V/P)**
Ambient ring $\mathbb{Z}_2$	3-manifold $M \simeq \mathrm{Spec}(\mathcal{O}_k)$	$\mathrm{Spec}(\mathcal{O}_k)$	**(V)**
Collatz support S_k shrinking toward attractor	Knot complement $X_K = M \setminus V_K^\circ$	$\mathrm{Spec}(\mathcal{O}_k) \setminus \{\mathfrak{p}\}$	**(V/P)**
Parity branching $B(n) = \mathbf{1}_{n \text{ odd}}$	Meridian $\alpha \in \pi_1(\partial V_K)$	Inertia generator $\tau \in I_\mathfrak{p}$	**(V/P)**
T_3-step direction (even/odd choice)	Longitude $\beta \in \pi_1(\partial V_K)$	Frobenius $\sigma \in \pi_1(\mathrm{Spec}(\mathbb{F}_\mathfrak{p}))$	**(V/P)**
Monodromy of Collatz paths in $\hat{\mathbb{Z}}$	Knot group $G_K = \pi_1(S^3 \setminus K)$	Galois group G_S with restricted ramification	**(V/P)**
v_2-step under T_3	Linking number $\mathrm{lk}(K, L)$	Legendre symbol $\left(\frac{p}{q}\right)$	**(V/P)**
Char. polynomial of v_2-transfer; supermartingale generating function	Alexander polynomial $\Delta_K(t)$	Iwasawa polynomial $f_K(T)$	**(V/P)**
Multi-step v_2 correlations	Milnor $\bar{\mu}$-invariants	Multiple residue symbols / Massey products	**(V/P)**
Attractor $\{1, 2\}$	Unknot; $G_K = \mathbb{Z}$, $\Delta_K = 1$	Unramified extension; $\pi_1(\mathrm{Spec}(\mathbb{Z})) = 1$	**(V/P)**
Orzechowski threshold $M_2 = 4$	Minimal 4-piece non-Arch. paradoxical decomposition	Ramification boundary	**(P)**
Transfer operator $\mu \mapsto (T_3)_*\mu$	Deck transformation; covering monodromy	Iwasawa-algebra action on class group	**(P)**

Remark 4.1. *The fundamental topology–arithmetic correspondences — $S^1 \leftrightarrow \mathrm{Spec}(\mathbb{F}_q)$, knot $\leftrightarrow$ prime, meridian $\leftrightarrow$ inertia τ, longitude $\leftrightarrow$ Frobenius σ — appear as Morishita's eq. (1.2) [4]. The Legendre/linking and Alexander/Iwasawa correspondences are Sections 2 and 7 of the same survey. The proposed identification of $B(n)$ with the inertia monodromy τ rather than Frobenius is interpretive: $B(n)$ heuristically corresponds to τ in the sense that it detects whether the trajectory passes through the ramified (odd) branch, while the Frobenius σ corresponds to the direction of the step itself. Making this precise is part of OP1.*

4.3 The Abelian Collatz Holonomy

Definition 4.2 (Abelian Collatz holonomy)**.** *Let $\gamma_n = (n, T_3(n), \ldots, T_3^{|\gamma_n|}(n))$ be a finite Collatz trajectory terminating at the attractor, where $|\gamma_n|$ denotes the number of steps. Define*

$$\begin{aligned}
\mathrm{hol}_{\mathrm{add}}(\gamma_n) &= \sum_{j=0}^{|\gamma_n|-1} v_2(\gamma_n^{(j)}) \in \mathbb{Z}_{\geq 0}, \\
\mathrm{hol}_{\mathrm{norm}}(\gamma_n) &= \frac{\mathrm{hol}_{\mathrm{add}}(\gamma_n)}{|\gamma_n|} \in \mathbb{Q}_{\geq 0}, \\
\mathrm{hol}_{\log}(\gamma_n) &= \mathrm{hol}_{\mathrm{add}}(\gamma_n) \cdot \log 2 \in \mathbb{R}_{\geq 0}.
\end{aligned}$$

In Kim's framework, holonomy proper is a group-valued path functional taking values in a p-adic Lie group; the additive form $\mathrm{hol}_{\mathrm{add}}$ is a candidate scalar shadow of an underlying $\mathbb{Z}_p$-valued holonomy in the abelianized setting. The lift to Kim's framework requires specification of the basepoint in $\pi_1^{\mathrm{un}}(\mathrm{Spec}(\mathbb{Z}_2))$ and the target p-adic Lie group; this is taken up in subsequent work and is recorded as an open problem in Section 7.

5 The Orzechowski Boundary as Ramification Boundary

The image-diversity threshold $M_2 = 4$ marks the boundary at which the Collatz ensemble loses the structural richness required for a non-Archimedean paradoxical decomposition in Orzechowski's sense [10]. We propose that this boundary corresponds, via the $\mathrm{M}^2\mathrm{KR}$ dictionary, to the ramification boundary in arithmetic topology:

- *Above the threshold* ($M_2 \geq 4$): the trajectory is in a regime analogous to a ramified extension, carrying nontrivial holonomy.
- *Below the threshold* ($M_2 < 4$): the trajectory approaches the vacuum, analogous to an unramified extension.
- *The attractor* $\{1, 2\}$: analogous to the unknot, corresponding to the unique unramified extension of $\mathbb{Q}$ (which has none).

The empirical characterization of this boundary in Section 6 shows that the crossing is a sharply localized late-trajectory event with a measurable phase transition in $\bar{v}_2$, supporting the proposed identification.

6 Empirical Results

All results in this section were produced by `collatz_arithmetic_probe.py` (CC0, available at [11]).

6.1 Supermartingale Sweep

Across $N \in \{50, 100, 200, 500, 1000, 2000\}$:

N	steps	$\sum_k \varepsilon_k$	violations	max ratio	min ratio	mean ratio
50	69	2.637	31	1.500	0.714	1.012
100	74	2.248	36	1.440	0.788	1.009
200	78	1.394	36	1.447	0.788	1.005
500	90	1.687	42	1.333	0.679	1.007
1000	112	2.651	53	1.482	0.583	1.008
2000	114	1.255	61	1.286	0.804	1.003

The Cesàro mean C_k is non-increasing across all tested N. Upward step-ratio violations exist at every scale but the mean ratio is uniformly within $[1.003, 1.012]$ of unity, and $\sum_k \varepsilon_k < 2.7$ throughout, well below $\mathbb{E}_{\mu_0}[f] \approx 0.5$–$0.7$. This supports Conjecture 2.1 in its Cesàro form. Stability of the normalization across $N \to \infty$ remains an open empirical question.

6.2 Abelian Holonomy on Small Trajectories

The quantity $\mathrm{hol}_{\mathrm{add}}(\gamma_n)$ was computed for $n = 1, \ldots, 100$. The principal structural findings are:

- *Approximately linear in length:* Pearson correlation of $\mathrm{hol}_{\mathrm{add}}$ with trajectory length is $r = 0.959$. The quantity reflects properties of the trajectory as a whole, not the starting point alone.
- *Weak dependence on $v_2(n)$:* Correlation of $\mathrm{hol}_{\mathrm{add}}$ with $v_2(n)$ is $r = -0.243$. The starting 2-adic depth has limited predictive power.
- *Normalized mean near $\mathbb{E}[v_2]$:* The mean of $\mathrm{hol}_{\mathrm{norm}}$ over $n \leq 100$ is 1.22, against the uniform-bit prediction $\mathbb{E}[v_2] = 1$. The excess of 0.22 reflects the non-uniform 2-adic digit distribution along Collatz trajectories.
- *Parity:* $\mathrm{hol}_{\mathrm{add}} \equiv v_2(n) \pmod 2$ in 55% of cases.

Selected values:

n	$v_2(n)$	length	$\mathrm{hol}_{\mathrm{add}}$	$\mathrm{hol}_{\mathrm{norm}}$
1	0	1	0	0.000
2	1	1	1	1.000
8	3	3	6	2.000
16	4	4	10	2.500
27	0	70	47	0.671

The famously long trajectory at $n = 27$ has the lowest normalized holonomy in the sample, consistent with approximate linearity in length rather than dependence on arithmetic depth of the start point.

6.3 Holonomy at the Vacuum

The attractor cycle $\{1, 2\}$ has canonical signature

$$v_2 \text{ sequence: } [0, 1, 0], \qquad \mathrm{hol}_{\mathrm{add}} = 1, \qquad \mathrm{hol}_{\mathrm{norm}} = \tfrac{1}{3}.$$

The ratio $2^{v_2(2)}/2^{v_2(1)} = 2$ matches the vacuum signature $M_4 = 2$ of [11]: the vacuum carries an irreducible 2-adic asymmetry intrinsic to the attractor.

Tail holonomy (last K steps before the attractor, computed for $n = 3, \ldots, 500$):

K	mean	std	min	max
5	1.500	0.395	1.200	3.000
10	1.424	0.593	1.000	4.500
20	1.185	0.180	0.900	2.650

Tail holonomy converges as K grows: the $K = 20$ window has standard deviation 0.18, substantially tighter than the full-trajectory distribution ($\sigma = 0.61$). The vacuum holonomy is therefore a narrow band around ≈ 1.18 rather than a single universal value.

6.4 The Orzechowski Crossing

For $N = 200$ (78 collapse steps), the threshold $M_2 = 4$ is crossed at step 75, i.e. at 96.2% of total collapse time. State at the crossing:

Quantity	**Value at crossing**
Support size	4
$\bar{v}_2$	0.750
σ_{v_2}	0.829
M_1 (conditional gap)	0.625
M_3 (branch balance)	1.000
M_4 ($\mathbb{E}[f\|B{=}0]/\mathbb{E}[f\|B{=}1]$)	0.375
M_5 (between-branch variance)	0.098

The v_2-mean shows a sharp drop at the crossing: pre-crossing window mean 1.424, post-crossing window mean 0.944, $\Delta\bar{v}_2 = -0.480$ (symmetric windows of five steps on either side of the crossing). The mean continues to fall toward the attractor value 0.5.

Observation 6.1. *The Orzechowski crossing is a late-trajectory event (last $\approx 4\%$ of steps) coinciding with a phase transition in $\bar{v}_2$. The trajectory spends $\approx 96\%$ of its lifetime above the non-Archimedean paradoxical-decomposition threshold, then collapses rapidly through the boundary to the attractor.*

The coincidence of the support-size-4 crossing with Orzechowski's four-piece minimum supports the proposed identification of this boundary with the ramification boundary on the arithmetic side.

7 Open Problems

The following problems remain open. They are listed in order of logical dependence; problems marked *[analytic]* require pen-and-paper proof, those marked *[structural]* require identification of a precise correspondence in an existing framework.

OP1. *[structural]* **Verification of the translation table.** For each row of the table in Section 4.2, provide a precise mathematical statement of the proposed correspondence using Morishita [5] and Kim [6], and identify any row requiring modification. The parity-versus-inertia identification is a particular priority.

OP2. *[structural]* **Lifting the scalar shadow to Kim's framework.** Specify (i) the correct basepoint in $\pi_1^{\mathrm{un}}(\mathrm{Spec}(\mathbb{Z}_2))$, (ii) the target p-adic Lie group, and (iii) the sense in which

discrete Collatz steps define morphisms in the étale fundamental groupoid, in such a way that the resulting holonomy reduces to $\mathrm{hol}_{\mathrm{add}}$ in the abelian quotient and is consistent with the empirical constraints $\mathrm{hol}_{\mathrm{norm}}(\{1,2\}) = 1/3$ and tail-holonomy convergence to $\approx 1.18 \pm 0.18$.

OP3. *[analytic]* **Proof of the supermartingale conjecture.** The Cesàro mean C_k is empirically non-increasing across $N \in \{50, \ldots, 2000\}$ with $\sum_k \varepsilon_k < 2.7$ throughout. A proof of Conjecture 2.1 is the primary analytic problem and the prerequisite for optional-stopping arguments on the holonomy side.

OP4. *[structural]* **Identification of $\mathrm{hol}_{\mathrm{norm}} = 1/3$.** The vacuum value $\mathrm{hol}_{\mathrm{norm}}(\{1,2\}) = 1/3$ is canonical. Identify the arithmetic object to which it corresponds under the $\mathrm{M}^2\mathrm{KR}$ dictionary.

OP5. *[structural]* **Formalization of the Orzechowski boundary.** Prove that the support-size-4 crossing corresponds to a specific change in the ramification data of the arithmetic path, and determine whether Orzechowski's four-piece minimality reflects a structural property of Kim's arithmetic Chern–Simons functional.

OP6. *[structural]* **Non-abelian extension.** Determine whether the non-abelian version of Kim's theory is accessible via Collatz trajectories, and whether the resulting invariant is a candidate arithmetic analog of a quantum knot invariant.

OP7. *[analytic]* **Bidirectional consequence.** Determine whether well-definedness and convergence of the lifted holonomy implies the Collatz conjecture, or a conditional version of it, via optional stopping applied to the holonomy supermartingale.

8 Summary

The Collatz map, viewed as a dynamical system on $\mathbb{Z}_2$ via Siegel's conjugacy, generates a natural family of paths in p-adic arithmetic space. The $\mathrm{M}^2\mathrm{KR}$ dictionary relates such paths to knot invariants and arithmetic topology, and Kim's arithmetic Chern–Simons theory provides the cohomological machinery to assign invariants to them. The empirical 2-adic transfer-operator framework provides the probabilistic structure on which a candidate scalar shadow of an abelian holonomy can be defined and computed.

The empirical results reported here — approximate linearity of $\mathrm{hol}_{\mathrm{add}}$ in trajectory length ($r = 0.96$), narrow-band convergence of the tail holonomy (1.18 ± 0.18), a robust supermartingale signature across six scales of N, and the Orzechowski crossing as a sharply localized late-trajectory phase transition at support size 4 with $\Delta\bar{v}_2 = -0.48$ — constitute measured properties of the system that constrain any precise lift of the framework to Kim's setting. The remaining work, collected in Section 7, is the verification of the proposed correspondences against the existing literature and the analytic proofs required to make the framework fully precise.

References

[1] B. Mazur, *Remarks on the Alexander polynomial*, unpublished manuscript, 1964.

[2] M. M. Kapranov, *Analogies between the Langlands correspondence and topological quantum field theory*, in *Functional Analysis on the Eve of the 21st Century*, Vol. 1, Progr. Math. **131**, Birkhäuser, 1995, pp. 119–151.

[3] A. Reznikov, *Three-manifolds class field theory*, Selecta Math. (N.S.) **3** (1997), 361–399.

[4] M. Morishita, *Analogies between knots and primes, 3-manifolds and number rings*, arXiv:0904.3399, 2009.

[5] M. Morishita, *Knots and Primes: An Introduction to Arithmetic Topology*, Universitext, Springer, 2012.

[6] M. Kim, *Arithmetic gauge theory: A brief introduction*, Modern Phys. Lett. A **33** (2018), no. 29.

[7] H.-J. Chung, D. Kim, M. Kim, J. Park, H. Yoo, *Entanglement entropies in the abelian arithmetic Chern–Simons theory*, Comm. Number Theory Phys. **18** (2024), 579–609.

[8] M. Carlson, H.-J. Chung, D. Kim, M. Kim, J. Park, H. Yoo, *Path integrals and p-adic L-functions*, Bull. London Math. Soc. **56** (2024), 1951–1966.

[9] M. Siegel, *A (p, q)-adic spectral framework for Collatz-type maps*, preprint.

[10] K. Orzechowski, *Paradoxical decompositions of finite-dimensional non-Archimedean normed spaces*, arXiv:2506.01528, 2025.

[11] JGPTech, *2-adic expectation dynamics in the Collatz map: an empirical study via transfer operators*, 2026. `https://github.com/JGPTech/Fun`